

Tessera 8

An eight-voice analogue polysynth with a digital heart and nothing in the way.

Tessera 8 pairs eight fully discrete analogue voices with a modern control core: a 49-key keybed with polyphonic aftertouch, a sixteen-slot modulation matrix, and a stereo effects engine that stays out of the signal path until you ask for it. Every parameter has its own control; nothing is more than one gesture away.

8	2+1	24 dB	9.4 kg
ANALOGUE VOICES	VCOS + SUB PER VOICE	OTA LADDER FILTER	WEIGHT

Discrete voice cards

Each voice is a removable card with its own calibration memory. Swap a card in under a minute without retuning the others.

Polyphonic aftertouch

Per-key pressure routes to any destination in the matrix. Full MPE over USB and five-pin MIDI.

Quiet by design

Linear power supply, balanced outputs, and a noise floor below −96 dBu at the main outputs.

Technical specifications

VOICE ARCHITECTURE

Polyphony	8 voices, analogue
Voice modes	Poly, Unison (1–8), Mono, Split
Oscillators	2 × VCO per voice
Waveforms	Saw, pulse (PWM), triangle
Sub oscillator	Square, −1 / −2 oct
Noise	White, pink
Oscillator sync	Hard, OSC 2 → OSC 1
Cross-modulation	Linear FM, 0–100%
Tuning stability	±0.5 cent, 0–40 °C

FILTER & AMPLIFIER

Type	4-pole OTA ladder, low-pass
Slope	24 dB/oct, switchable 12 dB
Cutoff range	16 Hz – 22 kHz
Resonance	To self-oscillation
Keyboard tracking	0–120%, continuous
VCA	Discrete, linear/exp response
Drive	Pre-filter, 0–18 dB

MODULATION

Envelopes	2 × ADSR, 1 ms – 30 s
LFOs	2 per voice + 1 global
LFO shapes	Sine, tri, saw, square, S&H, smooth random
LFO rate	0.01 Hz – 400 Hz, tempo sync
Matrix	16 slots, bipolar depth
Sources	26, incl. poly AT, velocity, MPE slide

EFFECTS

Engine	32-bit float, 48 kHz
Chorus	2- and 4-voice ensemble
Delay	Up to 2.5 s, tempo sync
Reverb	Plate and hall algorithms
Bypass	True analogue bypass relay

CONTROLS & KEYBOARD

Keybed	49 keys, semi-weighted
Aftertouch	Polyphonic and channel
Wheels	Pitch (spring) · Mod
Display	2.4″ OLED, 256 × 64
Controls	52 pots, 4 faders, 30 buttons
Presets	512 (256 factory)

CONNECTIVITY

Audio out	2 × ¼″ TRS balanced
Headphones	¼″ TRS, 140 mW @ 32 Ω
MIDI	5-pin DIN in / out / thru
USB	USB-C (class compliant), USB-A host
CV / gate	2 × 3.5 mm, 1 V/oct
Pedals	Sustain, expression

AUDIO PERFORMANCE

Output level	+16 dBu max, balanced
Noise floor	< −96 dBu (A-weighted)
Frequency response	10 Hz – 30 kHz, ±0.5 dB

POWER & PHYSICAL

Power	100–240 V AC, 50/60 Hz, 45 W
Supply	Internal linear PSU
Dimensions	812 × 318 × 96 mm
Weight	9.4 kg
Operating temp.	0 – 40 °C

Specifications subject to change. All values typical unless stated.

Signal flow

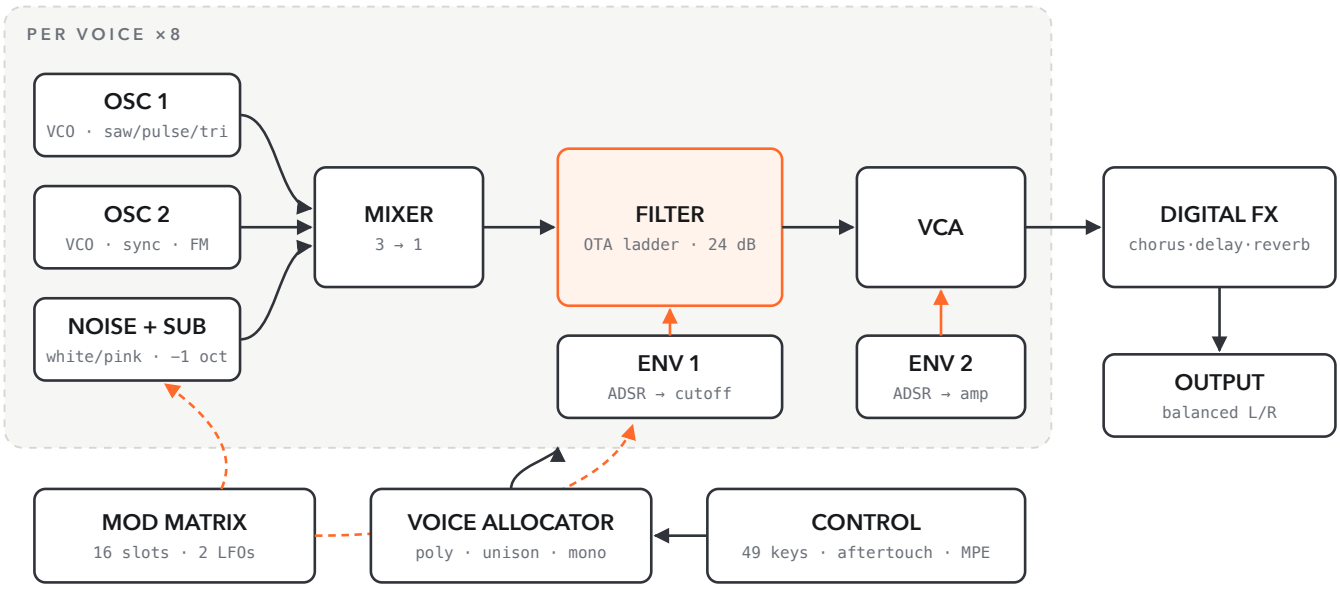


Figure 1 — Voice architecture. Black paths carry audio; orange paths carry modulation. Everything inside the dashed boundary is repeated on each of the eight voice cards.

THE VOICE

Each voice begins with two discrete VCOs built around matched transistor cores, with a sub-oscillator and a noise source alongside. A three-input mixer feeds the filter with roughly 6 dB of headroom before soft clipping, so turning everything up adds weight rather than distortion — until you add drive on purpose.

The filter is a four-pole ladder built from operational transconductance amplifiers. It can be switched to two-pole for a brighter, less resonant character, and it self-oscillates cleanly enough to be played as a sine source with keyboard tracking at 100%.

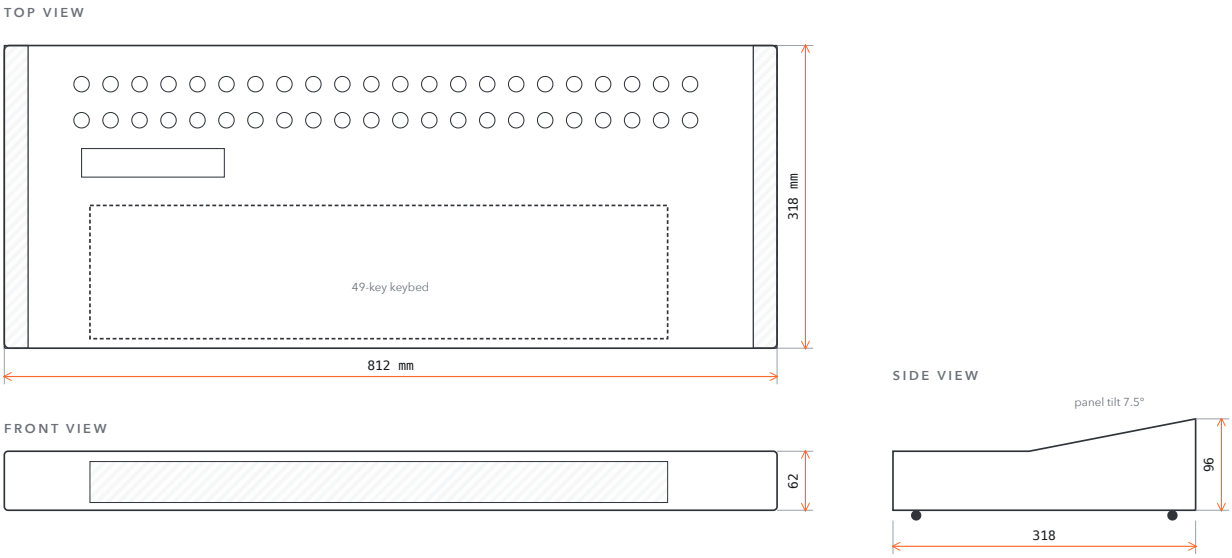
THE CONTROL CORE

A dedicated microcontroller per voice card scans and calibrates its own oscillators at power-on and every ten minutes thereafter, so the instrument stays in tune as it warms up without interrupting a performance.

The main processor runs the modulation matrix at 2 kHz per voice, the effects engine, and the voice allocator. When the effects are bypassed, a relay removes the converters from the signal path entirely: what reaches the outputs is the analogue voice and nothing else.

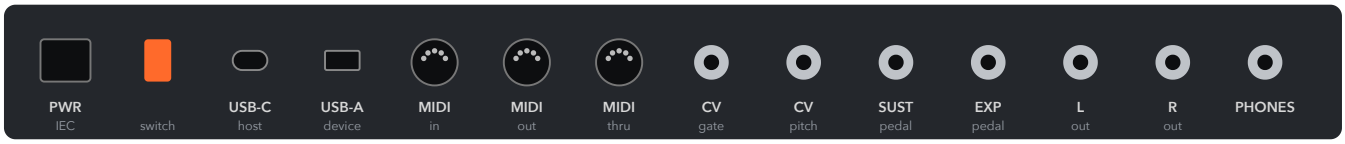
Calibration in the field. Voice cards store their own calibration tables. A card moved to another Tessera 8 carries its tuning with it; a new card calibrates itself in about forty seconds on first power-up.

Dimensions



All dimensions in millimetres. Height includes rubber feet. Drawing not to scale between views.

Rear panel



Left to right as seen from behind the instrument.

Ordering information

PART NO.	DESCRIPTION	INCLUDED
HI-T8-49	Tessera 8, 49-key, walnut end cheeks	IEC power cable, USB-C cable, quick-start card
HI-T8-49-BK	Tessera 8, 49-key, black ash end cheeks	As above
HI-VC-1	Spare voice card	Card, extraction tool
HI-CASE-T8	Padded transport case	–
HI-STAND-X	Folding X-stand, height adjustable 640–980 mm	–